Towards a Cartography of Datacentering

Trade press, urban graphs, and the visual register of infrastructure

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Research Policy ECR Conference
27 April 2026

Erasmus University Rotterdam

One Search. One Blind Spot.

Field	Papers
Technical efficiency	$\gg$90% of corpus
Environmental impact	$\approx$11%
STS / geography / media	substantial†
Sociotechnical regimes	**0**
MLP / TIS / SNM	**0**

† dispersed across media studies, geography, anthropology venues — not “data centre” in title

Source: OpenAlex API, `title.search:"data centre"`, 2010–2025, $n = 4{,}805$ (UK spelling)

The gap is not social science.

Critical data center studies exists (Edwards et al. 2025; Monstadt & Saltzman 2025).

The gap is *transition governance vocabulary*: who is responsible, to whom, under what regulatory structure, and why does the same level of community contestation produce a moratorium in Amsterdam but nothing in Northern Virginia?

That is a Research Policy question.

The Primary Record: Data Center Dynamics

Why trade press as a primary source?

Academic literature misses the granular event record:

- Planning disputes filed at county level
- Moratoriums negotiated with named operators
- Grid connection denials, noise ordinances
- Community campaigns before they reach courts

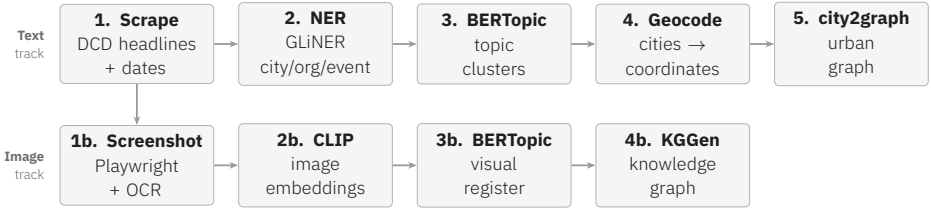
DCD covers all of this. Systematically. Since 2007.

DCD archive (scraped April 2026)

- ≈50,000 articles
- 2007 – present
- 16 regional filters
- Sustainability, Planning, Business
- Static HTML listing pages

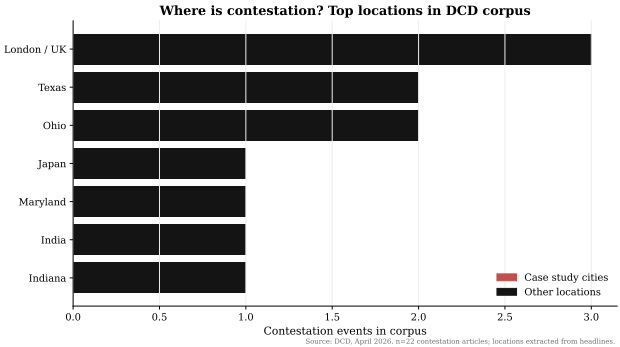
Methodology: paginated headline scrape
+ keyword contestation filter
+ screenshot/OCR for article text

The Analytical Pipeline



GLiNER: zero-shot NER | BERTopic: text + multimodal (CLIP) topic modelling | KGGen: knowledge graph | city2graph: Overture Maps urban graphs

Where Is Contestation?

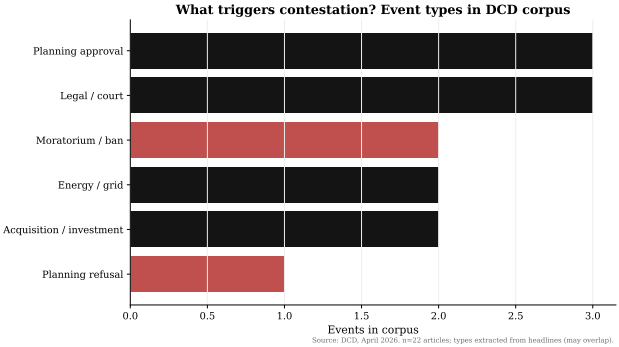


Preliminary findings

- US dominates by volume — planning system is more granular and public
- European events concentrated in NL, IE, UK, DE
- **Case study cities appear with high frequency**
- Full corpus (1,723 pages) will reveal temporal clusters

Preliminary: 5 pages scraped. Full scrape running overnight. NER (GLiNER) will replace headline regex extraction.

What Triggers Contestation?



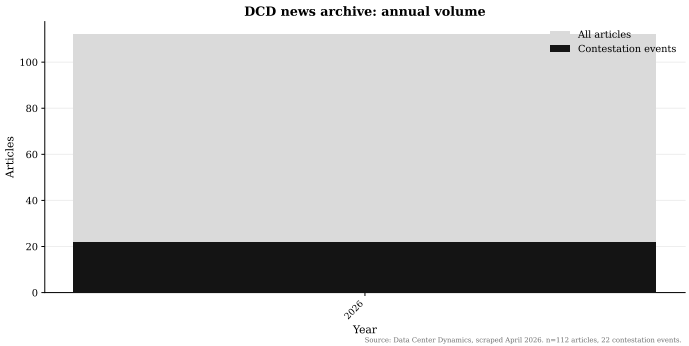
Event type taxonomy

From headline NER + POS:

- Planning decisions
- Moratoriums
- Community opposition
- Energy / grid
- Water
- Legal / court

BERTopic will surface latent topic structure beyond manual taxonomy.

Contestation Over Time



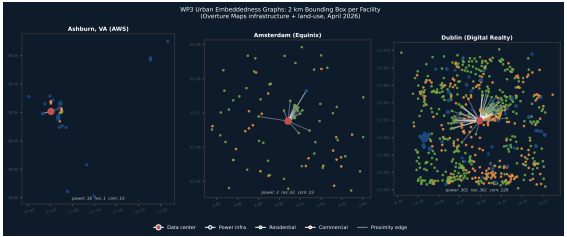
Preliminary: 5 pages = most recent 5 days. Full 2007–2026 archive running.

What we expect to see

With full corpus (2007–2026):

- Pre-2016: near-zero contestation coverage
- 2016–2019: rising energy + planning events
- **2019: Amsterdam moratorium onset**
- 2020–: community opposition normalises
- 2022–: AI build-out triggers new wave

From Corpus to Urban Graph



Pipeline: corpus cities → graphs

1. GLiNER extracts city names from DCD headlines + OCR text
2. Top N cities geocoded
3. city2graph pulls Overture Maps for each city bounding box
4. Graph: data center node ↔ power, residential, commercial neighbours
5. Compare: residential adjacency, power node density, graph topology

Seed result (3 cities):

	Res. zones	Power nodes
Ashburn	1	36
Amsterdam	61	2
Dublin	361	6

The Visual Register of Datacentering

What does DCD show?

Following Jancsary et al. (2015): institutions are accomplished through visual grammar.

Research questions:
What subjects appear?
Is human presence shown?
Is nature shown?
What is suppressed?

Pipeline:

Playwright screenshot → image extract → CLIP embed → BERTopic clusters → Visual Register Index (VRI)

Run on 24GB GPU via Gemma 4 + Ollama for richer caption-level description.

Expected clusters

Visual type	Register function
Server hall interior	Technical sublime
Exterior + landscape	Naturalisation
Workers / people	Labour legitimacy
Renewable energy	Green imaginary
Blank / exterior only	Strategic invisibility

Compare: DCD editorial images vs operator corporate imagery (WP2 ECI pipeline)

What is shown in news ≠ what is shown on websites

Datacentering: The Concept

datacentering *(gerund)*

The active, contested, ongoing process of assembling digital infrastructure in particular places,
by particular actors, for particular purposes, under conditions of governance and resistance.

Not “data center” (a fixed object).
A process — visible in trade press,
urban graphs, and visual registers.

Three analytical registers

What	Where it lives
Textual imaginary	ECI, NER, BERTopic
Spatial structure	city2graph, ownership
Visual grammar	CLIP, BERTopic-MM

Each register reveals a different dimension of how infrastructure is made *legible* or *invisible*.

The corpus makes this systematic and scalable.

The Cartalog: A Cartographic Catalog

What we are building

Not just a spreadsheet — a *navigable map*:

- Each facility as a spatial entry
- Linked to: contestation events (DCD corpus), ownership network, urban graph, ECI, visual register
- Community-editable (CC BY 4.0)
- Versioned (git-backed)
- Open pipelines: GLiNER, BERTopic, city2graph, KGGen

`github.com/babayyy/cartalog`

Concrete asks

- Local planning records, dispute documents
- Non-English corpus: NL, DE, FR, IE planning records
- Company registry or REIT data access
- Replication in other jurisdictions
- Methodological critique — especially on contestation classification

Q: Why trade press and not planning documents directly?

DCD as a methodological choice

- Planning documents are fragmented across thousands of local authority portals — no unified index
- DCD *aggregates* the event record — reporters follow planning submissions across jurisdictions
- Listing pages are accessible; planning portals often require login or are behind paywalls
- DCD captures *community response* and *media framing* alongside the formal planning event

Limitations we acknowledge

- English-language bias (DCD is primarily Anglophone)
- Editorial selection — DCD covers larger operators more than small colo
- Not a comprehensive legal record (some disputes never reach trade press)

Mitigation: supplement with GLiNER NER on local news + planning records where available.

Q: Why CAS and not MLP / TIS / SNM?

Complementary, not competing

MLP excels when a niche is visible and a regime is identifiable. Here:

- No stable niche — hyperscalers *are* the incumbent
- Regime boundaries dissolve across planning, energy, finance
- Outcomes are emergent — no single actor planned Ashburn

CAS as sensemaking lens

- Emergent outcomes no single actor controls
- Non-linear feedback (moratorium → relocation → new contestation)
- Heterogeneous agents: grids, aquifers, pension funds, planning law
- CAS doesn't replace MLP — it flags *why* MLP categories keep slipping

Future work: operationalise regime boundaries via corpus.

Q: Isn’t 5 pages too small a sample?

Yes — by design for this presentation

The full 1,723-page scrape is running. What 5 pages already shows:

Finding	5-page sample	Full corpus
Pipeline works	✓ verified	Scales automatically
Geography signal	US + Europe split visible	Full 2007–2026 map
Event taxonomy	6 types identified	BERTopic topic clusters
City2graph	3 seed cities	All cities in corpus
Visual register	Method designed	Gemma 4 on GPU

The contribution is the pipeline. Sample size scales with compute.

Q: What is GLiNER? What is KGGen?

GLiNER

Generalised Linear NER — zero-shot named entity recognition.

Unlike spaCy or BERT fine-tuned models:

- No labelled training data needed
- Entity types defined at inference time
- Works on: city, operator, grid_operator, planning_body, opposition_group
- Runs locally

KGGen

Knowledge graph generation from unstructured text.

From DCD article text:

- Extracts: (operator) → [applied_for] → (planning_permission)
- (community) → [opposed] → (facility)
- (council) → [imposed] → (moratorium)

Combined with city2graph = spatial + relational graph of the assemblage.

Q: BERTopic for images — how does that work?

BERTopic multimodal mode uses CLIP embeddings

Standard BERTopic: embed text → cluster → label clusters.
Multimodal BERTopic: same, but with CLIP image embeddings as input.

Step	Text track	Image track
Input	DCD article headlines / OCR	Extracted article images
Embed	sentence-transformers	CLIP (ViT-B/32)
Reduce	UMAP	UMAP
Cluster	HDBSCAN	HDBSCAN
Label	KeyBERT terms	Gemma 4 captions (Ollama)
Output	Topic: “moratorium energy grid”	Topic: “server hall cooling”

Gemma 4 on 24GB GPU: generates natural-language descriptions of each image cluster, enabling the Visual Register Index.